



Australian
National
University

Venue

Student Number

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Research School of Economics

PRACTICE EXAMINATION

Semester 1 – End of Semester, 2026

EMET2007/4007/6007_Semester 1 Econometric Methods

This paper is for ANU students only.

Examination Duration: 120 minutes

Reading Time: 0 Minutes

Exam Conditions:

N/A

Materials Permitted In The Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

No materials permitted

Materials To Be Supplied To Students:

1 x 20 page

Instructions To Students:

- There are 3 exam questions, each counting 20 raw marks.
- Answer all exam questions.
- Provide careful, complete, and self-contained answers.
- Do not merely state results. You are expected to show your work!
- Good luck!

Question 1

You study the effect of media consumption on cognitive development (as measured by a standardized math score) for Australian teenagers. Media consumption (for example: watching TV, using social media, Playstation) may affect a child's learning. You investigate the research question: Is media consumption negatively associated with math scores?

You have available sample data on test score, media consumption, and other demographic characteristics for 8,764 Australian teenagers for the year 2018. The data are drawn from the Longitudinal Survey of Australian Children (LSAC).

Here a brief description of some of the variables:

- `mathscr`: respondent's score on a standardized math test (between 0 and 600)
- `mediahrs`: respondent's average daily media consumption in hours
- `male`: dummy equal 1 if respondent identifies as male

Use the following Python output (partially edited) to answer the questions below.

PYTHON CODE AND OUTPUT (EDITED)

```
> formula = 'mathscr ~ mediahrs * male'
> reg = smf.ols(formula, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> print(reg.summary())
```

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept      531.2458      11.868      44.763      <2e-16      507.985      554.507
mediahrs       -2.2392      xxxxxx      xxxxxx      0.050      xxxxxx      xxxxxx
male           -0.0232       0.008      -2.942      0.003      -0.039      -0.008
mediahrs:male   -1.2766       0.967      -1.324      0.187      -3.172       0.619
=====
```

- [3 marks] Interpret and discuss the coefficient estimate for `male`. Is it statistically significant?
- [5 marks] Determine the t-statistic for `mediahrs`.
- [5 marks] Interpret and discuss the coefficient estimate for the interaction term.
- [7 marks] Consider your estimate of the effect of media consumption on math scores. Give an example of an omitted variable that could bias the above results. If this variable would be included, how would you expect your estimates to change? Explain.

Question 2

Are the following statements true or false? Provide a brief and complete explanation. (Note: you will not receive any credit without providing a correct explanation.)

(a) [5 marks]

Statement:

In the multiple regression model, omitting an explanatory variable may not necessarily lead to omitted variables bias.

(b) [5 marks]

Statement:

In autoregressions, allowing for additional lags (that is, increasing p in the $AR(p)$ model) increases R^2 .

(c) [5 marks]

Statement:

Given a random sample $Y_1, \dots, Y_{10}$, the following two estimators are equally good for estimating $E(Y_7)$:

- the simple average of Y_1 and Y_{10} ;
- $2/3 \cdot Y_6 + 1/3 \cdot Y_9$.

(d) [5 marks]

Statement: In the simple regression model, if $\beta_1 = 0$, then $\hat{\beta}_0 = \bar{Y}$.

Question 3

Consider the linear regression model

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + u_i, \quad \beta_1 \neq 0.$$

(a) [10 marks]

Suppose that $\beta_2 = 0$. Define and derive the OLS estimator of β_1 . Prove that this estimator is unbiased. Explicitly state any additional assumptions that you may need to establish unbiasedness.

(b) [10 marks]

Now suppose that $\beta_2 \neq 0$. Under which circumstances is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n X_{1i} Y_i}{\sum_{i=1}^n X_{1i}^2}?$$

Provide a complete proof.

End of Exam Questions
